EXERCISE – IMPLEMENTATION OF K-MEANS ALGORITHM

AIM

To implement the K-Means clustering algorithm using scikit-learn to group data points (e.g., Iris flower samples) into a specified number of clusters based on feature similarity, and visualize the resulting clusters.

PROCEDURE

1. Import the required libraries: scikit-learn for loading the dataset and the KMeans algorithm, and Matplotlib for visualization.
2. Load the Iris dataset, which contains measurements of three species of Iris flowers.
3. Extract the feature matrix (X) containing the four numerical attributes of each sample.
4. Create a KMeans model with a chosen number of clusters (n_clusters=3) and a fixed random state for reproducibility.
5. Fit the KMeans model to the data so that it learns the cluster centers and assigns each sample to a cluster.
6. Retrieve the cluster labels assigned to each data point.
7. Visualize the clusters by plotting two selected features, coloring points according to their assigned cluster labels.
8. Run the program and observe the scatter plot of the formed clusters.

PROGRAM

```
from sklearn.datasets import load_iris
from sklearn.cluster import KMeans
import matplotlib.pyplot as plt

# Load data
data = load_iris()
X = data.data

# Apply KMeans
kmeans = KMeans(n_clusters=3, random_state=0)
kmeans.fit(X)
labels = kmeans.labels_

# Plot (2 features only for visualization)
plt.scatter(X[:, 0], X[:, 1], c=labels, cmap='rainbow')
plt.xlabel("Feature 1")
plt.ylabel("Feature 2")
plt.title("K-Means Clustering")
plt.show()
```

OUTPUT

A scatter plot is displayed showing the Iris data points colored according to the three clusters discovered by the K-Means algorithm. Points belonging to the same cluster share the same color. The plot uses the first two features (sepal length and sepal width) for visualization.

(Note: The exact visual appearance of the plot depends on the runtime environment. The algorithm successfully partitions the data into three distinct clusters corresponding approximately to the three Iris species.)

Result: The K-Means clustering algorithm was successfully implemented on the Iris dataset. The data points were grouped into three clusters based on feature similarity, and the resulting clusters were visualized using a scatter plot.